Complex Tori and Elliptic Curves

Why "draw a line and reflect" on a cubic is secretly addition on a torus

A self-contained route from one course in complex analysis to the group law on an elliptic curve

A Bizarre Rule — and Why It Is Secretly Arithmetic

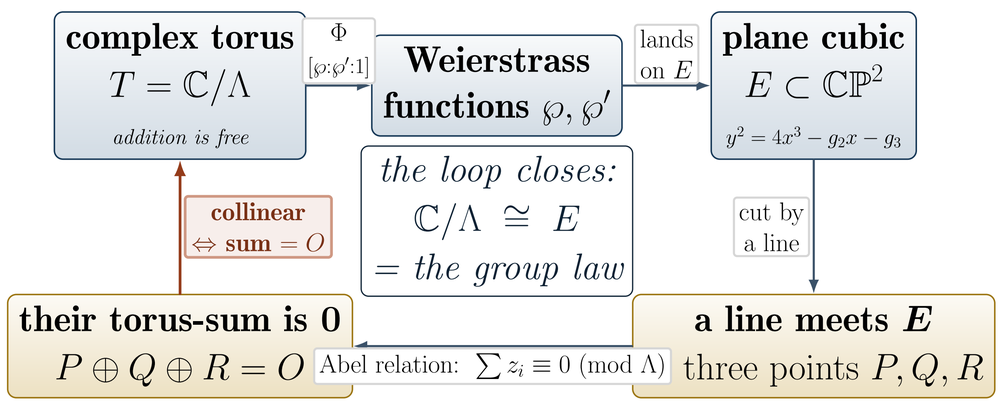
On a smooth cubic curve there is a strange way to "add" two points P and Q: *draw the straight line through them, find the third point R where it meets the curve, then reflect R across the x-axis.* Call the result $P \oplus Q$.

Astonishingly, this rule is commutative *and* associative — it turns the points of the curve into an abelian group. **Why should "draw a line and reflect" obey the laws of arithmetic? And why, of all things, the reflection?**

This poster answers both. The cubic only *looks* like a curve in the plane; it is secretly a *torus* $\mathbb{C}/\Lambda$ — the complex plane rolled up by a lattice. On the torus, $\oplus$ is nothing but addition of complex numbers, z mod Λ, which is obviously a group. The Weierstrass functions $\wp, \wp'$ *draw* the torus as the cubic, and carry ordinary addition exactly onto "line, third point, reflect."

Main theorem. $\Phi([z]) = [\wp(z) : \wp'(z) : 1]$, with $0 \mapsto O = [0{:}1{:}0]$, is an isomorphism $\mathbb{C}/\Lambda \xrightarrow{\sim} E$ of *both* Riemann surfaces *and* groups. Every feature of the rule, the reflection included, is then forced.

The Argument at a Glance



The torus has an effortless addition. The bridge Φ carries it to the cubic. A *line* meets the cubic in three points whose torus-sum is 0 — and that single loop *is* the chord–tangent law. The three columns below walk it in order:
(1–2) the two characters — torus and elliptic functions;
(3–4) the bridge — the Weierstrass $\wp$ and the embedding;
(5–6) the payoff — "line, then reflect," and why one object wears three faces.

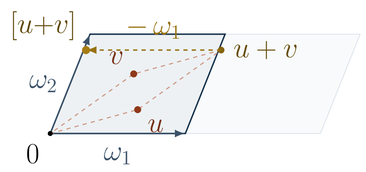
1. The torus $\mathbb{C}/\Lambda$: addition you already know

A *lattice* $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ is a grid of parallelograms tiling the plane $\mathbb{C}$. Glue the opposite edges of one tile and you get a **torus** (a donut surface) $T = \mathbb{C}/\Lambda$: a point of T is just a complex number "read modulo the grid."

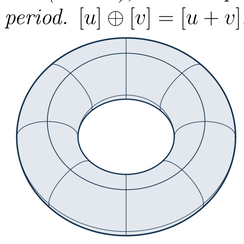
Its addition is free. Add the two complex numbers and wrap around:

$$[u] \oplus [v] \;=\; [u+v].$$

This inherits everything from $\mathbb{C}$: commutative and associative, identity $[0]$, inverse $[-z]$. *No geometry, no effort — the torus is already a perfect abelian group.*



To add: form $u+v$ in $\mathbb{C}$ (dashed), then wrap back into the tile by a period. $[u] \oplus [v] = [u+v]$.



The same tiling, glued into the torus surface — the group we transport.

2. Elliptic functions: rigid by Liouville

To compare the torus with a curve we need *functions on the torus* — functions on $\mathbb{C}$ that repeat on the grid, $f(z+\omega) = f(z)$. These are **elliptic functions**, and they are astonishingly rigid.

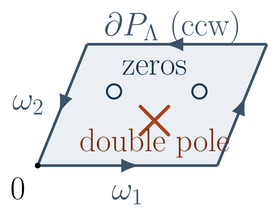
Why rigid? A doubly periodic function with no poles is bounded, so by Liouville's theorem it is **constant**. Hence any non-constant one *must* have poles — and integrating around one tile (opposite sides cancel) shows it has as many poles as zeros. The very simplest possible has a single double pole.

The seed. Even the *positions* are pinned down: integrating $z\,f'(z)/f(z)$ around a tile gives the **Abel relation**

$$\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda.$$

This one fact is what forces the reflection step at the very end.

The function realising the simplest case — one double pole, nothing else — is the *Weierstrass* $\wp$, built next.



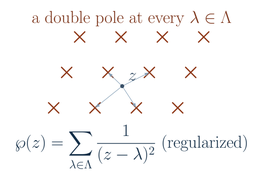
Integrate f, f'/f, $z\,f'/f$ counterclockwise around one tile ∂P_Λ. Opposite sides carry equal values (f repeats), so they cancel — forcing $\#\text{zeros} = \#\text{poles}$ and the Abel relation $\sum(\text{zeros}) - \sum(\text{poles}) \in \Lambda$.

3. The Weierstrass function $\wp$: the simplest one

To *build* an elliptic function with a single double pole, sum $1/z^2$ over the whole lattice (with constants for convergence):

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus 0} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

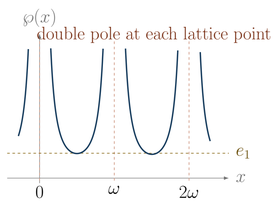
It is even, one double pole per tile; its derivative $\wp'$ is odd, with a triple pole.



The sum runs over the whole lattice, planting a double pole at every λ; near z the four closest terms dominate.

One magic identity. Near 0, $\wp = z^{-2} + 3G_4 z^2 + \cdots$, so $\wp'^2$ and $4\wp^3$ *both* begin with $4z^{-6}$. Subtract them and the right multiples of $\wp$ and 1: every pole cancels, leaving a bounded — hence constant — function. So $\wp$ and $\wp'$ satisfy one cubic relation:

$$\boxed{\wp'^2 = 4\wp^3 - g_2\,\wp - g_3} \qquad g_2 = 60G_4, \quad g_3 = 140G_6.$$



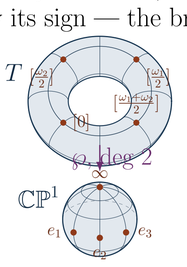
On the real axis $\wp$ has a double pole at every lattice point and one valley per period, bottoming at a real root $e_1 = \wp(\omega/2)$ — the picture behind "a single double pole."

4. The bridge Φ: drawing the torus as a cubic

Now send each torus point to the projective point

$$\Phi(z) = [\wp(z) : \wp'(z) : 1], \qquad \Phi(0) = O = [0{:}1{:}0].$$

The boxed identity puts the image on the cubic $E : y^2 = 4x^3 - g_2x - g_3$, and in fact Φ is a biholomorphism $\mathbb{C}/\Lambda \xrightarrow{\sim} E$. *Why a bijection?* $\wp$ is 2-to-1 (it is even), and $\wp'$ tells the two preimages apart by its sign — the branched cover below.



The map $\wp : T \to \mathbb{CP}^1$ is a $2:1$ branched cover with 4 branch values: three roots e_1, e_2, e_3 and ∞. Riemann–Hurwitz: $0 = 2{\cdot}2 - 4{\cdot}(2-1)$.

$\wp : T \to \mathbb{CP}^1$ folds the torus 2-to-1 onto the sphere, branching only over the three roots e_1, e_2, e_3 and ∞. Recording $\wp'$ as well unfolds it into the cubic E — a flawless dictionary $T \cong E$.

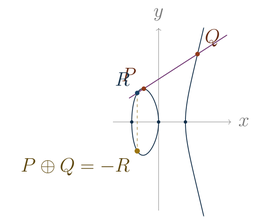
Now the payoff. We have a perfect dictionary $T \cong E$. *Transport the torus's effortless addition through Φ* — what does it become on the curve?

5. The payoff: addition becomes "line, then reflect"

Here is the one fact behind it all. A line $ax + by + c = 0$ pulls back to $a\wp + b\wp' + c$, an elliptic function with a *single triple pole* at O — so exactly three zeros, which (by Abel) sum to 0 on the torus:

$$P, Q, R \text{ collinear on } E \iff P \oplus Q \oplus R = O.$$

Thus $P \oplus Q = \ominus R$. And $\ominus$ on the torus is $z \mapsto -z$, i.e. $(x, y) \mapsto (x, -y)$ ($\wp$ even, $\wp'$ odd) — **reflection across the x-axis**. Mystery solved.



The line through P and Q meets E at a third point R; reflecting R across the x-axis gives $P \oplus Q$. Collinear means torus-sum 0; reflection is $z \mapsto -z$.

6. Why it matters: one object, three faces

What we built is a single abelian group seen three ways — and the isomorphisms between those views *are* the subject:

$$\underbrace{\mathbb{C}/\Lambda}_{\text{analysis}} \;\cong\; \underbrace{\mathrm{Pic}^0(E)}_{\text{geometry}} \;\cong\; \underbrace{E(k)}_{\text{arithmetic}} .$$

Three faces. **Analytic** — $\mathbb{C}/\Lambda$: complex numbers added modulo a lattice.
Geometric — $\mathrm{Pic}^0(E)$: degree-0 divisor classes (line bundles); Abel–Jacobi makes E its own Jacobian.
Arithmetic — $E(k)$: the points themselves, under chord–tangent, over *any* field k.

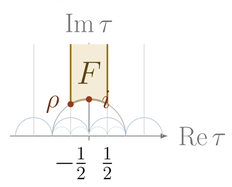
That last face is the doorway to modern number theory: because the law is *rational*, it travels to every field at once.

Over $\mathbb{Q}$. Mordell–Weil: $E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$ is finitely generated. The *rank* r is deep — Birch–Swinnerton-Dyer (a Clay Millennium Prize problem) predicts it from the order of vanishing of the L-function $L(E, s)$ at $s = 1$.

Over $\mathbb{F}_p$. Hasse's bound $\left|\#E(\mathbb{F}_p) - (p+1)\right| \le 2\sqrt{p}$ pins the point count, and the hardness of *reversing* the group law underwrites elliptic-curve cryptography.

Galois & modularity. The torsion assembles into a representation $\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{Z}_\ell)$; every $E/\mathbb{Q}$ is *modular* — the theorem behind Wiles's proof of Fermat's Last Theorem.

And $j = 1728\, g_2^3/\Delta$ names the curve inside the moduli space of *all* elliptic curves — the next picture:



F is one representative per torus, so $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ is the moduli space of complex tori — coordinate the j-invariant; i, ρ are the extra-symmetric tori.

Author: ⟨Your Name⟩ • ⟨email⟩ • seeking PhD positions in arithmetic & algebraic geometry.

From the textbook *Complex Tori and Elliptic Curves* (self-contained; ∼80 solved exercises).

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